Outer Measures, Pre-Measures, and Product Measures

Andrew Hwang*

Abstract

We have gone over the development of the Lebesgue measure by beginning with elementary notions of length and volume and extending them to more complicated sets through coverings and outer measure. In these notes we revisit that construction from a more general perspective. We introduce the Carathéodory Extension Theorem, which recovers countable additivity on a distinguished σ-algebra, and the Hahn–Kolmogorov Extension Theorem, which builds an outer measure from a premeasure on an algebra; together these are the ways we generate measures. Using these extension theorems we explore Lebesgue–Stieltjes measures and how different choices of increasing right-continuous functions give rise to different measures on the Borel σ-algebra. As an application we examine the Cantor set, which is uncountable yet has measure zero, showing that cardinality and measure are independent notions of size. From here, we also explore product spaces and their respective product measures, along with the theory of sections, leading into the Fubini–Tonelli theorems.

Contents

*Notes from a talk given at Columbia Summer Learning Seminar, 08/18/2026.

1 Goal

We know how to measure intervals and boxes on the real line using the Lebesgue measure. But there are many other types of measures. The idea of this talk is to discuss how we construct general measures beyond the Lebesgue measure.

1.1 Recall: σ-Algebras and Measures

Definition 1.1 (σ-algebra)**.** A collection $\mathcal{M}$ of subsets of a set X is a *σ-algebra* if:

1. $\emptyset \in \mathcal{M}$ (and hence $X \in \mathcal{M}$);
2. $A \in \mathcal{M} \implies A^c \in \mathcal{M}$ (closed under complements);
3. $A_1, A_2, \ldots \in \mathcal{M} \implies \bigcup_{j=1}^\infty A_j \in \mathcal{M}$ (closed under countable unions).

Definition 1.2 (Measure)**.** A function $\mu : \mathcal{M} \to [0, \infty]$ is a *measure* if:

1. $\mu(\emptyset) = 0$;
2. for any countable collection $\{A_j\}_{j=1}^\infty \subset \mathcal{M}$ of *pairwise disjoint* sets,

$$\mu\left(\bigcup_{j=1}^\infty A_j\right) = \sum_{j=1}^\infty \mu(A_j)$$

(countable additivity).

2 Outer Measures

An *outer measure* is a function

$$\mu^* : 2^X \longrightarrow [0, \infty]$$

satisfying:

1. $\mu^*(\emptyset) = 0$;
2. if $A \subseteq B$, then $\mu^*(A) \leq \mu^*(B)$;
3. given a countable collection of sets $\{A_i\}_{i=1}^{\infty}$, we have that

$$\mu^* \left(\bigcup_{i=1}^{\infty} A_i \right) \;\leq\; \sum_{i=1}^{\infty} \mu^*(A_i).$$

Note the two ways in which this is weaker than a measure: the domain is all of the power set rather than a σ-algebra, and countable additivity has been relaxed to countable *sub*additivity. The first is what makes outer measures easy to build; the second is the price we pay, and the rest of this section is about recovering additivity on a suitable subfamily of sets.

2.1 Carathéodory Measurability

Definition 2.1 (Carathéodory measurable)**.** We say that a set $A \in \mathcal{P}(X)$ is *Carathéodory measurable* (or just *μ^*-measurable*) if for all $E \in \mathcal{P}(X)$ we have that

$$\mu^*(E) \;=\; \mu^*(E \cap A) + \mu^*(E \setminus A).$$

The condition says that A splits every test set E additively. Subadditivity already gives $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \setminus A)$ for free, so the content of the definition is always the reverse inequality.

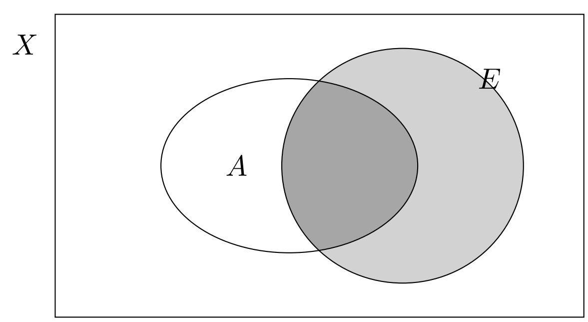

Figure 1: Carathéodory's splitting condition: A cuts the test set E into $E \cap A$ and $E \setminus A$, and the two outer measures must add back up to $\mu^*(E)$.

Example 2.2 (Null sets)**.** Given some space X and a null set E (that is, $\mu^*(E) = 0$), we can see that for some arbitrary $B \in \mathcal{P}(X)$ we have $\mu^*(B \cap E) + \mu^*(B \setminus E) \geq \mu^*(B)$ because of countable subadditivity. Moreover, we can see that since $B \cap E \subseteq E$, by monotonicity $\mu^*(B \cap E) \leq \mu^*(E) = 0$. Similarly, we can say that $B \setminus E \subseteq B$, so $\mu^*(B \setminus E) \leq \mu^*(B)$. Hence we have

$$\mu^*(B \cap E) + \mu^*(B \setminus E) \;\leq\; 0 + \mu^*(B) \;=\; \mu^*(B).$$

Therefore $\mu^*(B \cap E) + \mu^*(B \setminus E) = \mu^*(B)$, so every null set is Carathéodory measurable.

2.2 The Measurable Sets Form a σ-Algebra

Let $\mathcal{B} = \{A \in \mathcal{P}(X) : A \text{ is } \mu^*\text{-measurable}\}$. Then $\mathcal{B}$ is a σ-algebra on X.

- **$\emptyset, X \in \mathcal{B}$:** $\mu^*(E \cap \emptyset) + \mu^*(E \setminus \emptyset) = 0 + \mu^*(E)$.
- **Closed under complements:** the condition $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A)$ is symmetric in A and A^c, since $E \cap A^c = E \setminus A$ and $E \setminus A^c = E \cap A$. So $A \in \mathcal{B} \iff A^c \in \mathcal{B}$, for free.
- **Closed under countable unions:** apply the definition twice to get $A \cup B \in \mathcal{B}$, then disjointify $\bigcup_{i=1}^{\infty} A_i$ and pass to the limit.

2.3 Carathéodory's Extension Theorem

Theorem 2.3 (Carathéodory)**.** *Given an outer measure $\mu^* : 2^X \to [0, \infty]$ and $\mathcal{B} = \{E : E \in \mathcal{P}(X) \text{ and } E \text{ is } \mu^*\text{-measurable}\}$, the restriction of μ^* onto $\mathcal{B}$ is a measure and $\mathcal{B}$ is a σ-algebra. That is, $\mu^*|_{\mathcal{B}} = \mu$ and $\mu(A) = \mu^*(A)$ if $A \in \mathcal{B}$.*

The main results are:

- $\mathcal{B}$ is a σ-algebra;
- $\mu^*|_{\mathcal{B}}$ is a measure;
- starting from an outer measure defined on the power set, we can restrict the domain to create a measure.

3 Pre-Measures and the Hahn–Kolmogorov Theorem

Carathéodory's theorem takes an outer measure as input. The next question is where outer measures come from. The answer is that they can be manufactured by covering, starting from a set function defined only on a much smaller and more concrete family of sets.

3.1 Algebras (Boolean Algebras)

Definition 3.1 (Algebra)**.** Given a set X, an *algebra* $\mathcal{A}$ on X is a collection of subsets of X such that

- $X \in \mathcal{A}$;
- if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$, where the complement is taken relative to X;
- if $\{A_i\}_{i=1}^{n} \subseteq \mathcal{A}$, then
$$\bigcup_{i=1}^{n} A_i \in \mathcal{A}.$$

So an algebra is a σ-algebra with countable unions weakened to finite unions.

3.2 Pre-Measures

Definition 3.2 (Pre-measure)**.** Given a set X and an algebra $\mathcal{A}$ on X, a *premeasure* is a function

$$\mu_0 : \mathcal{A} \to [0, \infty]$$

such that

- $\mu_0(\emptyset) = 0$;
- if $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ is a countable collection of pairwise disjoint sets and

  $$\bigcup_{i=1}^{\infty} A_i \in \mathcal{A},$$

  then

  $$\mu_0\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu_0(A_i).$$

The countable additivity requirement is conditional: it is imposed only when the countable union happens to land back inside $\mathcal{A}$, which for an algebra is not automatic.

3.3 The Hahn–Kolmogorov Theorem

Theorem 3.3 (Hahn–Kolmogorov)**.** *Let $\mathcal{A} \subseteq \mathcal{P}(X)$ be an algebra and $\mu_0 : \mathcal{A} \to [0, \infty]$ a premeasure. For $E \in \mathcal{P}(X)$, define*

$$\mu^*(E) = \inf \left\{ \sum_{i=1}^{\infty} \mu_0(A_i) \ : \ A_i \in \mathcal{A},\ E \subseteq \bigcup_{i=1}^{\infty} A_i \right\}.$$

Then μ^* is an outer measure on X.

Then apply Carathéodory. Let $\mathcal{B} = \{A \in \mathcal{P}(X) : A \text{ is } \mu^*\text{-measurable}\}$. Then $\mathcal{B}$ is a σ-algebra and $\mu^*|_{\mathcal{B}}$ is a measure.

3.4 Where the Resulting Measure Lives

The two theorems chain together. One shows that $\mathcal{A} \subseteq \mathcal{B}$; then

$$\mathcal{A} \subseteq \mathcal{B} \quad \overset{\mathcal{B} \text{ is a } \sigma\text{-algebra}}{\Longrightarrow} \quad \sigma(\mathcal{A}) \subseteq \mathcal{B} \quad \overset{\mu^*|_{\mathcal{B}} \text{ is a measure}}{\Longrightarrow} \quad \mu^*|_{\sigma(\mathcal{A})} \text{ is a measure,}$$

and altogether

$$\mathcal{A} \subseteq \sigma(\mathcal{A}) \subseteq \mathcal{B} \subseteq \mathcal{P}(X).$$

3.5 Carathéodory vs. Hahn–Kolmogorov

The two inclusions $\sigma(\mathcal{A}) \subseteq \mathcal{B}$ are typically strict, and the gap is not a technicality. In the case of Lebesgue measure on $\mathbb{R}$, $\sigma(\mathcal{A})$ is the Borel σ-algebra while $\mathcal{B}$ is the strictly larger σ-algebra of Lebesgue measurable sets — the Carathéodory construction automatically includes every null set, whereas the Borel sets do not.

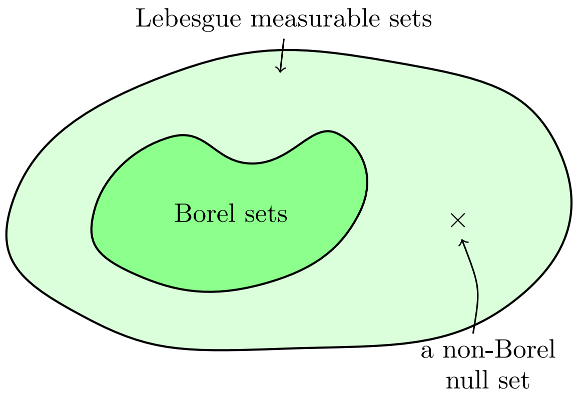

Figure 2: The Lebesgue measurable sets properly contain the Borel sets.

4 Borel and Lebesgue–Stieltjes Measures

Using what we have developed, we can now construct Borel measures and, specifically, Lebesgue–Stieltjes measures.

Let $F : \mathbb{R} \to \mathbb{R}$ be **increasing** and **right-continuous**.

Let $\mathcal{A}$ be the collection of finite disjoint unions of half-open intervals $(a, b]$, where $-\infty \leq a \leq b \leq \infty$. Then $\mathcal{A}$ is an algebra and

$$\sigma(\mathcal{A}) \;=\; \mathcal{B}_{\mathbb{R}}.$$

Define $\mu_0 : \mathcal{A} \to [0, \infty]$ by

$$\mu_0\big((a,b]\big) \;=\; F(b) - F(a), \qquad \mu_0\left(\bigsqcup_{i=1}^{n} (a_i, b_i]\right) \;=\; \sum_{i=1}^{n} \big(F(b_i) - F(a_i)\big).$$

4.1 Lebesgue–Stieltjes Measures

By Hahn–Kolmogorov, μ_0 extends to a measure μ_F on $\mathcal{B}_{\mathbb{R}}$ with

$$\mu_F\big((a,b]\big) \;=\; F(b) - F(a).$$

We call μ_F the **Lebesgue–Stieltjes measure** of F.

Taking $F(x) = x$:

$$\mu_F\big((a,b]\big) \;=\; b - a \;=\; \lambda\big((a,b]\big).$$

So Lebesgue measure is the Lebesgue–Stieltjes measure of the identity. Other choices of F recover other familiar measures:

$F(x) = x$	Lebesgue measure λ
$F = \mathbf{1}_{[0,\infty)}$	Dirac measure δ_0
$F(x) = \lfloor x \rfloor$	counting measure on $\mathbb{Z}$

5 An Application: The Cantor Set

5.1 The Question

We have now formally defined the construction of the Lebesgue measure as well as other Lebesgue–Stieltjes measures. From this, we have the following results under the standard Lebesgue measure m:

- $m(\mathbb{Q}) = 0$;
- $m((0,1)) = 1$;
- $m(\mathbb{R}) = \infty$;
- $m((0,1) \setminus \mathbb{Q}) = 1$.

We can see that the measure of uncountable sets like the interval $(0,1)$ or the reals is positive. Then, is the measure of uncountable sets always positive?

5.2 The Cantor Set

We construct the Cantor set $C \subset [0,1]$ by successively removing open middle thirds. Let

$$C_0 = [0,1].$$

Remove the open middle third to get

$$C_1 = \left[0, \tfrac{1}{3}\right] \cup \left[\tfrac{2}{3}, 1\right].$$

Remove the open middle third of each remaining interval to get

$$C_2 = \left[0, \tfrac{1}{9}\right] \cup \left[\tfrac{2}{9}, \tfrac{1}{3}\right] \cup \left[\tfrac{2}{3}, \tfrac{7}{9}\right] \cup \left[\tfrac{8}{9}, 1\right],$$

and so on: at stage n, C_n consists of 2^n closed intervals, each of length 3^{-n}. The Cantor set is defined as

$$C = \bigcap_{n=0}^{\infty} C_n.$$

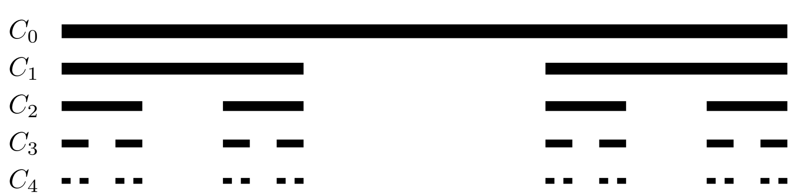

Figure 3: The first five stages of the middle-thirds construction.

Remark 5.1 (Measure of C)**.** Each C_n is a finite union of 2^n closed intervals of length 3^{-n}, so $m(C_n) = (2/3)^n$. Since $C \subseteq C_n$ for every n, monotonicity gives $m(C) \leq (2/3)^n \to 0$, and therefore $m(C) = 0$. Each C_n is closed, so C is closed as well.

5.3 Every Point of C is a Limit Point

Let $x \in C$. For each n, let I_n be the unique interval of C_n containing x, so

$$\text{length}(I_n) = 3^{-n}, \qquad I_0 \supset I_1 \supset I_2 \supset \cdots.$$

Let x_n be an endpoint of I_n with $x_n \neq x$. Endpoints of C_n-intervals are never removed later, so $x_n \in C$ for all n. Since $x_n, x \in I_n$,

$$|x_n - x| \;\leq\; \text{length}(I_n) \;=\; 3^{-n} \;\to\; 0,$$

so $x_n \to x$ with $x_n \neq x$. Hence x is a limit point of C.

Corollary 5.2. *C is uncountable. (In general, every nonempty perfect subset of a complete metric space is uncountable: no countable set can be perfect, since one can always find, inductively, a nested sequence of closed balls avoiding each listed point in turn, and by completeness their intersection is nonempty and contains a point not on the list.)*

Remark 5.3 (Cardinality vs. measure)**.** It is worth pausing on how strange this is. The Cantor set C satisfies

$$|C| = |\mathbb{R}| = |(0,1)|,$$

i.e. C has the same cardinality as the entire real line — it is "just as large" as $(0,1)$ in the sense of counting points. And yet

$$m(C) = 0,$$

while $m((0,1)) = 1$. So C is a set with the full cardinality of the continuum, but zero length. This shows that cardinality and Lebesgue measure are fundamentally different ways of measuring the "size" of a set: a set can be uncountable — even continuum-sized — while still being negligible from the point of view of measure.

6 Product Measures

Naturally, we don't work in just one space. For instance, consider any space with multiple coordinates, or joint distributions and their joint CDFs. These all require taking the product of multiple different measurable spaces.

Let us recall what the Cartesian product was. If we have two sets A and B, the Cartesian product is defined as follows:

$$A \times B := \{(a,b) : a \in A \text{ and } b \in B\}.$$

Consider the product space $\mathbb{R} \times \mathbb{R}$. This is one of the simplest and most common examples of a product space. More generally, we may consider a product of spaces

$$\prod_n X_n.$$

If we want to define a measure on such a product space, we must first specify a σ-algebra of measurable subsets of that space. This motivates the question: given σ-algebras on the individual spaces, how should we construct a natural σ-algebra on their product?

6.1 The Product σ-Algebra

A very natural way we might think about what a σ-algebra of a product space might be is the set of rectangles. For now, let us just think of a product space of two measurable spaces $(X, \mathcal{M})$ and $(Y, \mathcal{N})$. We define

$$\mathfrak{R} := \{A \times B : A \in \mathcal{M} \text{ and } B \in \mathcal{N}\}.$$

So $\mathfrak{R}$ seems like it would be a nice σ-algebra for our product space; but, is it really a σ-algebra?

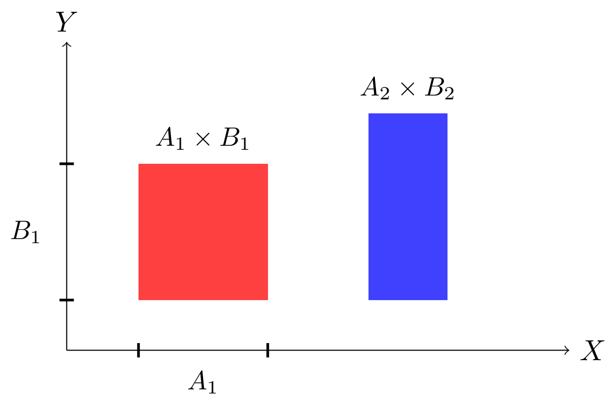

Figure 4: Measurable rectangles in $X \times Y$.

It turns out that $\mathfrak{R}$ is not a σ-algebra. Recall that one of the requirements of a σ-algebra is that it must be closed under complements. However, $\mathfrak{R}$ contains only rectangles of the form $A \times B$. In general, the complement of a rectangle is not itself a rectangle, so it need not belong to $\mathfrak{R}$. Thus, $\mathfrak{R}$ is not closed under complements and therefore is not a σ-algebra.

To fix this, we take the smallest σ-algebra containing all of the rectangles in $\mathfrak{R}$. That is, we define

$$\sigma(\mathfrak{R}),$$

the σ-algebra generated by $\mathfrak{R}$. This is what we call the **product σ-algebra**, and we write

$$\mathcal{M} \otimes \mathcal{N} := \sigma(\mathfrak{R}).$$

6.2 Projection Maps

Moreover, another very natural structure that arises from a product space is the notion of a **projection map**. For example, given the product space $X \times Y$, we have the two coordinate projections

$$\pi_1 : X \times Y \to X, \qquad \pi_1(x, y) = x,$$

and

$$\pi_2 : X \times Y \to Y, \qquad \pi_2(x, y) = y.$$

In other words, π_1 simply extracts the first coordinate of (x, y), while π_2 extracts the second coordinate. Since these projection maps arise naturally from the product space, we would like our product σ-algebra to make both π_1 and π_2 measurable.

Recall that a function

$$f : (S, \mathcal{A}) \to (T, \mathcal{B})$$

is measurable if, for every measurable set $E \in \mathcal{B}$, its preimage $f^{-1}(E)$ belongs to $\mathcal{A}$. Therefore, for the first projection π_1 to be measurable, we must have

$$\pi_1^{-1}(A)$$

measurable in $X \times Y$ for every $A \in \mathcal{M}$. Similarly, for the second projection π_2 to be measurable, we must have

$$\pi_2^{-1}(B)$$

measurable in $X \times Y$ for every $B \in \mathcal{N}$. Thus, the smallest σ-algebra on $X \times Y$ that makes both coordinate projections measurable is the σ-algebra generated by all of these preimages:

$$\sigma\left(\{\pi_1^{-1}(A) : A \in \mathcal{M}\} \cup \{\pi_2^{-1}(B) : B \in \mathcal{N}\}\right).$$

6.3 The Two Descriptions Agree

We will now show that this is exactly the same σ-algebra as the one generated by the measurable rectangles. That is,

$$\mathcal{M} \otimes \mathcal{N} \;=\; \sigma(\mathfrak{R}) \;=\; \sigma\left(\{\pi_1^{-1}(A) : A \in \mathcal{M}\} \cup \{\pi_2^{-1}(B) : B \in \mathcal{N}\}\right).$$

For any $A \in \mathcal{M}$ and $B \in \mathcal{N}$,

$$\pi_1^{-1}(A) = A \times Y, \qquad \pi_2^{-1}(B) = X \times B.$$

Thus, the preimages of the coordinate projections are themselves measurable rectangles. Conversely, every measurable rectangle can be written using these preimages:

$$A \times B = \pi_1^{-1}(A) \cap \pi_2^{-1}(B).$$

Therefore, the measurable rectangles and the projection preimages generate the same σ-algebra. Hence,

$$\boxed{\mathcal{M} \otimes \mathcal{N} \;=\; \sigma(\mathfrak{R}) \;=\; \sigma\left(\{\pi_1^{-1}(A) : A \in \mathcal{M}\} \cup \{\pi_2^{-1}(B) : B \in \mathcal{N}\}\right).}$$

6.4 The Algebra of Finite Disjoint Unions

Let $\mathcal{A}$ be the collection of *finite disjoint unions* of rectangles,

$$\mathcal{A} \;=\; \left\{ \bigsqcup_{i=1}^{n} (A_i \times B_i) \;:\; A_i \in \mathcal{M},\; B_i \in \mathcal{N} \right\}.$$

Then $\mathcal{A}$ is an algebra, and

$$\mathfrak{R} \;\subseteq\; \mathcal{A} \;\subseteq\; \sigma(\mathfrak{R}) \;=\; \sigma(\mathcal{A}) \;=\; \mathcal{M} \otimes \mathcal{N}.$$

Now apply "$\mathcal{C} \subseteq \sigma(\mathcal{D}) \implies \sigma(\mathcal{C}) \subseteq \sigma(\mathcal{D})$" in both directions:

- $\mathfrak{R} \subseteq \mathcal{A} \subseteq \sigma(\mathcal{A})$, so $\sigma(\mathfrak{R}) \subseteq \sigma(\mathcal{A})$;
- $\mathcal{A} \subseteq \sigma(\mathfrak{R})$, so $\sigma(\mathcal{A}) \subseteq \sigma(\mathfrak{R})$.

6.5 Constructing the Product Measure

Goal. The most natural requirement we want is that for a "clean" rectangle $A \times B$ with $A \in \mathcal{M}$, $B \in \mathcal{N}$, the product measure should satisfy

$$(\mu \times \nu)(A \times B) = \mu(A)\,\nu(B).$$

Step 1: The algebra of rectangles. Let $\mathcal{A}$ be the collection of all *finite disjoint unions* of rectangles $A \times B$ with $A \in \mathcal{M}$, $B \in \mathcal{N}$. One checks that $\mathcal{A}$ is an algebra: it is closed under complements and finite unions, and every $E \in \mathcal{A}$ can be written as

$$E = \bigsqcup_{j=1}^{n} A_j \times B_j$$

for some finite collection of pairwise disjoint rectangles $A_j \times B_j$.

Step 2: A premeasure on $\mathcal{A}$. Define $\pi : \mathcal{A} \to [0, \infty]$ by

$$\pi(E) = \sum_{j=1}^{n} \mu(A_j)\,\nu(B_j) \qquad \text{where } E = \bigsqcup_{j=1}^{n} A_j \times B_j.$$

Step 3: Hahn–Kolmogorov extension. Define an outer measure π^* on all of $X \times Y$ by

$$\pi^*(E) = \inf \left\{ \sum_{j=1}^{\infty} \pi(E_j) \;:\; E \subset \bigcup_{j=1}^{\infty} E_j,\ E_j \in \mathcal{A} \right\}.$$

By the Carathéodory extension theorem, the restriction of π^* to the σ-algebra $\mathcal{M} \otimes \mathcal{N}$ generated by $\mathcal{A}$ is a genuine measure, and it agrees with π on $\mathcal{A}$.

Definition 6.1 (Product measure)**.** We call this resulting measure the *product measure*, denoted $\mu \times \nu$, on $(X \times Y, \mathcal{M} \otimes \mathcal{N})$. In particular,

$$(\mu \times \nu)(A \times B) = \mu(A)\,\nu(B)$$

for all $A \in \mathcal{M}$, $B \in \mathcal{N}$, as desired.

6.6 Is It Well-Defined?

A set $E \in \mathcal{A}$ may be written as a finite disjoint union of rectangles in more than one way, so we must check that $\pi(E)$ does not depend on the decomposition chosen.

Proposition 6.2. *Suppose $A \times B$ is a rectangle that is a (finite or countable) disjoint union of rectangles $A_j \times B_j$. Then for $x \in X$ and $y \in Y$,*

$$\chi_A(x)\chi_B(y) = \chi_{A \times B}(x, y) = \sum_j \chi_{A_j \times B_j}(x, y) = \sum_j \chi_{A_j}(x)\chi_{B_j}(y).$$

Integrating both sides and applying the monotone convergence theorem to interchange the sum and the integrals:

$$\int_X \int_Y \chi_A(x)\chi_B(y)\, d\nu(y)\, d\mu(x) = \int_X \int_Y \sum_j \chi_{A_j}(x)\chi_{B_j}(y)\, d\nu(y)\, d\mu(x)$$

$$\int_X \chi_A(x) \left(\int_Y \chi_B(y)\, d\nu(y)\right) d\mu(x) = \int_X \sum_j \chi_{A_j}(x) \left(\int_Y \chi_{B_j}(y)\, d\nu(y)\right) d\mu(x)$$

$$\int_X \chi_A(x)\, \nu(B)\, d\mu(x) = \int_X \sum_j \chi_{A_j}(x)\, \nu(B_j)\, d\mu(x)$$

$$\nu(B) \int_X \chi_A(x)\, d\mu(x) = \sum_j \nu(B_j) \int_X \chi_{A_j}(x)\, d\mu(x)$$

and therefore

$$\boxed{\mu(A)\, \nu(B) = \sum_j \mu(A_j)\, \nu(B_j).}$$

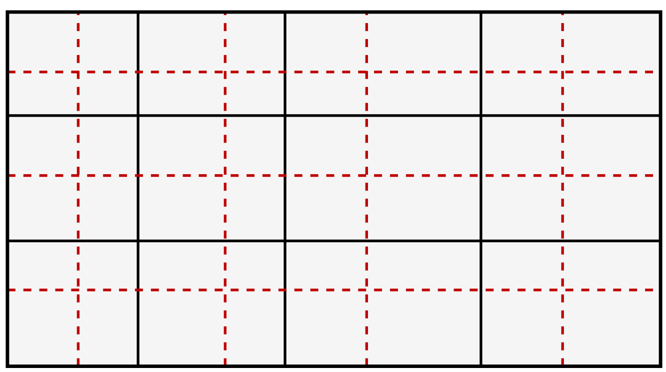

Figure 5: Two different decompositions of the same rectangle into disjoint sub-rectangles. Overlaying them gives a common refinement, on which both values of π can be computed and compared.

6.7 σ-Finiteness

Definition 6.3 (σ-finite)**.** A measure μ on a measurable space $(X, \mathcal{M})$ is *σ-finite* if there exists a countable collection of sets $\{E_j\}_{j=1}^{\infty} \subset \mathcal{M}$ such that

$$X = \bigcup_{j=1}^{\infty} E_j \qquad \text{and} \qquad \mu(E_j) < \infty \text{ for every } j.$$

In other words, X can be covered by countably many sets of finite measure, even if $\mu(X)$ itself is infinite.

Example 6.4. Lebesgue measure m on $\mathbb{R}$ is σ-finite: taking $E_j = (-j, j)$ for $j = 1, 2, 3, \dots$ gives

$$\mathbb{R} = \bigcup_{j=1}^{\infty} (-j, j), \qquad m((-j, j)) = 2j < \infty,$$

even though $m(\mathbb{R}) = \infty$.

7 Sections

7.1 Sections of Sets

Let X and Y be sets, and let $E \subset X \times Y$. For $x \in X$, the *x-section* of E is the subset of Y given by

$$E_x = \{y \in Y : (x, y) \in E\}.$$

Similarly, for $y \in Y$, the *y-section* of E is the subset of X given by

$$E^y = \{x \in X : (x, y) \in E\}.$$

Intuitively, E_x is the "slice" of E you get by fixing the first coordinate at x and looking at which y's pair with it; E^y is the analogous slice fixing the second coordinate.

Example 7.1. If $E = A \times B$ is a rectangle, then

$$E_x = \begin{cases} B & \text{if } x \in A \\ \emptyset & \text{if } x \notin A \end{cases} \qquad \text{and} \qquad E^y = \begin{cases} A & \text{if } y \in B \\ \emptyset & \text{if } y \notin B. \end{cases}$$

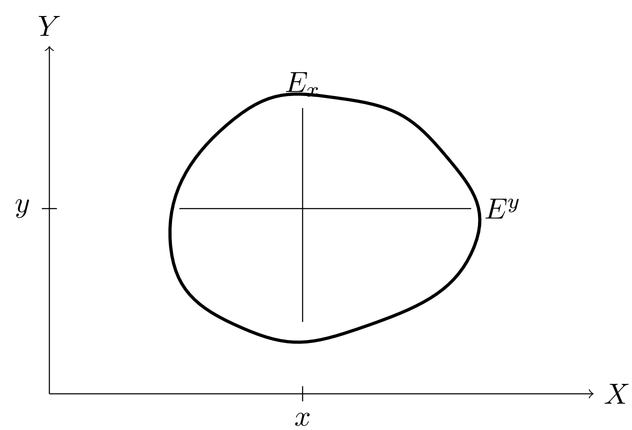

Figure 6: The x-section and y-section of a set $E \subset X \times Y$.

7.2 Sections of Functions

Let $f : X \times Y \to \overline{\mathbb{R}}$ (or $\mathbb{C}$) be a function. For $x \in X$, the *x-section* of f is the function $f_x : Y \to \overline{\mathbb{R}}$ defined by

$$f_x(y) = f(x, y).$$

Similarly, for $y \in Y$, the *y-section* of f is the function $f^y : X \to \overline{\mathbb{R}}$ defined by

$$f^y(x) = f(x, y).$$

In words, f_x is obtained by freezing the first coordinate at x and letting y vary, while f^y freezes the second coordinate and lets x vary. Note that

$$(f_x)(y) = (f^y)(x) = f(x, y)$$

for all $x \in X$, $y \in Y$ — they are just two different ways of slicing the same function.

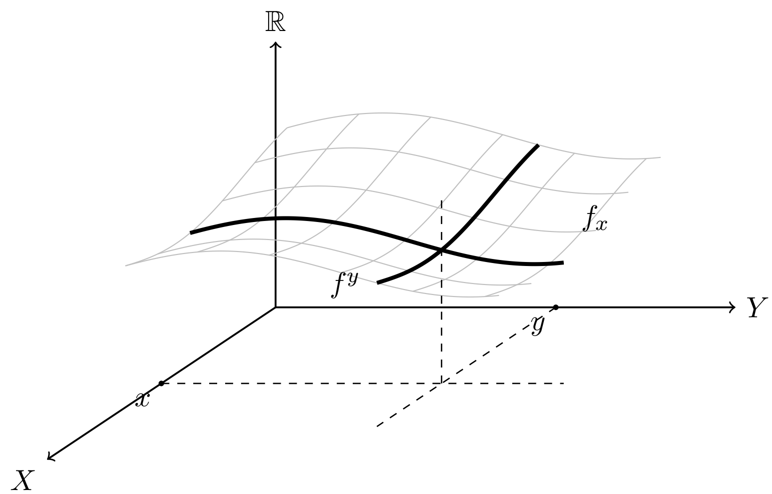

Figure 7: The graph of f over $X \times Y$, with the section f_x obtained by freezing the first coordinate and the section f^y obtained by freezing the second.

7.3 Sections and the Product Measure

Theorem 7.2. *Suppose $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are σ-finite measure spaces. If $E \in \mathcal{M} \otimes \mathcal{N}$, then the functions $x \mapsto \nu(E_x)$ and $y \mapsto \mu(E^y)$ are measurable on X and Y, respectively, and*

$$\mu \times \nu(E) \;=\; \int \nu(E_x)\, d\mu(x) \;=\; \int \mu(E^y)\, d\nu(y).$$

This is the statement that the measure of a set can be recovered by integrating the sizes of its slices, and it is the set-level precursor of the theorems in the next section.

8 Tonelli's and Fubini's Theorems

Theorem 8.1 (Tonelli)**.** *Suppose that $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are σ-finite measure spaces. If $f \in L^+(X \times Y)$, then*

$$\int f\, d(\mu \times \nu) \;=\; \int \left[\int f(x, y)\, d\nu(y) \right] d\mu(x) \;=\; \int \left[\int f(x, y)\, d\mu(x) \right] d\nu(y).$$

Theorem 8.2 (Fubini)**.** *Suppose that $(X,\mathcal{M},\mu)$ and $(Y,\mathcal{N},\nu)$ are σ-finite measure spaces. If $f \in L^1(\mu\times\nu)$, then $f_x \in L^1(\nu)$ for a.e. $x \in X$, and $f^y \in L^1(\mu)$ for a.e. $y \in Y$, and*

$$\int f\,d(\mu\times\nu) \;=\; \int\left[\int f(x,y)\,d\nu(y)\right]d\mu(x) \;=\; \int\left[\int f(x,y)\,d\mu(x)\right]d\nu(y).$$

The two theorems differ only in their hypothesis on f: Tonelli requires nonnegativity, Fubini requires absolute integrability. Neither hypothesis can simply be dropped, as the next section shows.

9 Why L^1 is Necessary: A Counterexample to Fubini

Define $f : [0,1]^2 \to \mathbb{R}$ by

$$f(x,y) = \frac{x^2-y^2}{(x^2+y^2)^2}, \qquad (x,y)\neq(0,0), \qquad f(0,0)=0.$$

The iterated integrals both exist, but disagree. One can compute

$$\int_0^1 f(x,y)\,dy = \frac{1}{1+x^2} \quad\Longrightarrow\quad \int_0^1\left(\int_0^1 f(x,y)\,dy\right)dx = \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4},$$

while

$$\int_0^1 f(x,y)\,dx = \frac{-1}{1+y^2} \quad\Longrightarrow\quad \int_0^1\left(\int_0^1 f(x,y)\,dx\right)dy = \int_0^1 \frac{-dy}{1+y^2} = -\frac{\pi}{4}.$$

So

$$\int_0^1\left(\int_0^1 f(x,y)\,dy\right)dx = \frac{\pi}{4} \;\neq\; -\frac{\pi}{4} = \int_0^1\left(\int_0^1 f(x,y)\,dx\right)dy.$$

Why Fubini doesn't apply. One can check that

$$\iint_{[0,1]^2} |f(x,y)|\,d(x,y) = \infty,$$

so $f \notin L^1([0,1]^2, m\times m)$. The hypothesis of Fubini's theorem fails, and indeed its conclusion fails: the two iterated integrals, though individually well-defined and finite, are not equal.

Moral. Absolute integrability over the product space is not a technicality — without it, the order of integration can genuinely matter.

References

[1] G. B. Folland, *Real Analysis: Modern Techniques and Their Applications*, 2nd ed., Wiley, 1999.

[2] T. Tao, *An Introduction to Measure Theory*, Graduate Studies in Mathematics, vol. 126, American Mathematical Society, 2011.